CFCs

Chlorofluorocarbons (CFCs) are chemicals made up of carbon, chlorine, and fluorine atoms. They are nontoxic, nonflammable, and extremely stable. Their low reactivity makes them very useful, but also highly destructive. They can survive for over 100 years, which gives them time to diffuse into the stratosphere. There, they are broken down by the intense UV light to release chlorine, which reacts with ozone to form oxygen.

First produced in 1928, CFCs were used as coolants for refrigerators, and later in a wide range of aerosol products, such as insect sprays, hair spray, and spray paints.

The replacements for CFCs were hydrochlorofluorocarbons (HCFCs), which also deplete the ozone layer, although to a much lesser extent, and hydrofluorocarbons (HFCs). HCFC production was banned in developed countries in 2020, and from 2030 will be banned in developing countries. HFCs do not harm the ozone layer—but are powerful greenhouse gases, and a phaseout has begun, with the aim of an 80 percent reduction by 2050.